

Independent Reviewer Guide

Suggested evaluation protocol

Review categories

A. Mathematical formulation

- Check action versus imposed background constraint.
- Verify signs, normalization, derivatives, conservation, and limiting behavior.
- Identify any equation that does not follow under the stated assumptions.

B. Numerical implementation

- Inspect solver tolerances, event handling, interpolation, and constraint monitoring.
- Confirm dual-solver independence.
- Check whether frozen parameters were changed between diagnostics.

C. Boltzmann and perturbations

- Inspect exact source-code changes.
- Verify gauge conventions and stress-energy source normalization.
- Test initial conditions and numerical stability.

D. Observational statistics

- Verify reference datasets, covariance matrices, window functions, degrees of freedom, and control model.
- Distinguish point comparisons from full likelihood inference.

E. Scientific claims

- Separate demonstrated internal consistency from empirical competitiveness and fundamental derivation.
- Flag wording that overstates independent verification or exclusion of alternatives.

Recommended conclusion labels

- Suitable for external technical discussion with minor corrections.
- Requires major revision before journal submission.
- Numerically interesting but not presently reproducible.
- Internally inconsistent for stated reason(s).
- Potentially publishable after independent reproduction and a full likelihood analysis.